AZ-305 — Exam-Day Cheat Sheet (v2, logic-first)

Exam: 2026-06-22 (Mon) · **Calibration:** 87–92% on recent mocks · Pass = 70%

How to use this sheet:

- Don't memorize. Walk the **decision trees** — they read the question for you.
- Mnemonics only appear for **arbitrary lists** (TM routing methods, the 6 Rs, Cosmos consistency levels, etc.) where there's no underlying logic.
- On exam morning: read it **once**, slowly. Then close it. No cramming.

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Universal exam-grading rules

These trump every decision tree below — read the question through this lens first.

- **"Minimize admin effort"** → simpler **setup**, not less ongoing work. Managed > self-managed. Built-in > custom. PaaS > IaaS.
- **"Minimize cost"** → pick the **smallest SKU that meets the requirement**. Don't overshoot. Reserved capacity / spot / autoscale / elastic pools all count as cost levers.
- **"Least privilege"** → built-in role > custom role; resource scope > RG scope > subscription scope > management group scope.
- **"Minimize downtime / disruption"** → online/live migration over offline. Replication-based DR over backup/restore.
- **Two answers seem right?** Pick the one that **matches the exact words** in the requirement.
- **"Most cost-effective"** ≠ **"cheapest"**. It means cheapest that still meets the SLA/perf requirement.
- **"Active-active" vs "active-passive"** is a hard constraint — if the question says "must serve from both regions" → active-active. If "secondary can take over within X minutes" → active-passive is fine.
- **"Recommend a solution / minimize effort"** — Microsoft loves managed services. Default to the most-managed option that fits.
- **First instinct usually wins.** Flag-and-return, don't second-guess unless you find a hard contradiction with the requirements.

D1 — Identity, Governance, Monitoring

🌳 Tree 1: Entra ID SKU selection

Question: "Which Entra ID license is required for feature X?"

```
What feature do they need?
├─ SSO to a SaaS app from Microsoft's gallery, basic MFA,
│  security defaults                                    → FREE
│
├─ CONDITIONAL ACCESS (rule-based: "if user in group X
│  and signing in from outside country Y, require MFA")  → P1
├─ Self-service password reset WITH on-prem writeback    → P1
├─ Application Proxy (publish on-prem app via Entra)     → P1
├─ Cloud Sync / Connect Sync advanced filters + writeback → P1
├─ Connect Health, password protection (banned passwords) → P1
├─ Dynamic groups / group-based licensing                 → P1
│
├─ PIM — Privileged Identity Management
│  (JIT activation, approval workflow, time-bound roles)  → P2
├─ Identity Protection (risk-based CA: risky user /
│  risky sign-in, ML-scored)                             → P2
│
├─ Modern Access Reviews, Entitlement Management,
│  Lifecycle Workflows, PIM for Groups                   → Entra ID Governance
│  (separate add-on SKU; sits on top of P1 or P2)
│
└─ External users / B2B / B2C identities                 → External ID
   (separate SKU; was "Azure AD External Identities")
```

Key confusion lock:

- **Conditional Access ≠ Identity Protection.**

    - CA = **rule-based**, you write the conditions (group / location / device / app). **P1.**
    - Identity Protection = **risk-based**, Microsoft's ML scores the user and sign-in. Then CA can consume the risk score as a condition. **P2.**
- **PIM** is for **elevated roles** (admin scope). **P2.**
- "Just-in-time role activation" → always PIM → always P2.
- "Sign-in risk policy" → Identity Protection → P2.
- "Require MFA when user is outside corporate network" → Conditional Access → P1 is enough.

🌳 Tree 2: Hybrid identity sync method

Question: "How should on-prem AD users sign in to Entra?"

```
Do they need ADFS-style federation (smart cards,
on-prem MFA, custom claims, third-party MFA)?
├─ YES → AD FS Federation (highest admin effort; only if forced)
│
└─ NO → They want SSO with minimal effort. Then:
   │
   Can the password hash live in the cloud?
   ├─ NO (compliance forbids it) → Pass-Through Authentication (PTA)
   │      • Needs ≥ 2 PTA agents on-prem for HA
   │      • Every sign-in hits on-prem (dependency on on-prem uptime)
   │
   └─ YES → Password Hash Sync (PHS) — default recommendation
          │
          Which sync agent?
          ├─ Modern, lightweight, multi-agent auto-HA, small forest        → CLOUD SYNC
          │  (greenfield, < 150k objects, no device writeback, no
          │   group writeback, no SQL filtering on attribute expressions)
          │
          └─ Legacy heavy install, advanced filtering, device writeback,   → CONNECT SYNC
             group writeback, large multi-forest, on-prem Exchange hybrid
             (HA pattern: 1 primary + 1 staging; staging is NOT HA, only
              promote-on-demand)
```

Admin-effort order (low → high): Cloud Sync < Connect Sync < PTA < Federation.

🌳 Tree 3: Authorization (RBAC / ABAC / custom)

```
Need to grant access?
├─ Built-in role fits exactly? → USE BUILT-IN. Always preferred.
│
├─ Built-in is too broad? → Add ABAC condition (e.g., "blob has tag = X")
│  to a built-in role assignment. Storage + Key Vault support ABAC.
│
└─ Built-in + ABAC still doesn't fit? → CUSTOM ROLE
   (only if you genuinely need a unique permission set)

Scope it as narrow as possible:
Resource > Resource Group > Subscription > Management Group
```

🌳 Tree 4: Governance (Policy / RBAC / Locks / Blueprints / Initiatives)

Goal	Tool
Restrict WHAT can be created (e.g., allowed SKUs, allowed regions, required tags)	**Azure Policy**
Restrict WHO can do something	**RBAC**
Prevent accidental delete/modify	**Resource Lock** (CanNotDelete / ReadOnly)
Bundle multiple policies as a set	**Policy Initiative** (definition set)
Deploy a complete environment (policies + RBAC + ARM templates + RGs together)	**Azure Blueprints** (being deprecated → use Deployment Stacks + Template Specs going forward, but on AZ-305 still answer Blueprints)
Force tag inheritance from RG to child resources	**Policy with `modify` effect + remediation task**
Enforce that something is deployed (e.g., diagnostic settings, agent)	**Policy with `deployIfNotExists` (DINE) + remediation task**

Trick: Policy assignment alone covers **new/updated** resources only. For **existing** resources to comply, you need a **remediation task**.

🌳 Tree 5: Monitoring agent + logs

```
Modern VM monitoring (Azure or Arc)? → AMA (Azure Monitor Agent) + DCR
                                      (Data Collection Rules)
                                      + assign via Azure Policy at scale
                                      + DCE (Data Collection Endpoint) if private link

MMA / OMS Agent → DEAD (retired Aug 2024). Never the answer in 2026.
Azure Diagnostics extension (WAD/LAD) → legacy, only for very specific
                                        metrics-to-storage scenarios.
```

Log Analytics retention tiers (for "keep logs for N years" questions):

Tier	Duration	Cost	Latency
Interactive	30 days (default) up to **2 years** (730 d) per table	$$$	Sub-second KQL
Archive	up to **12 years** per table (after interactive)	$	Hours (Search Job or Restore)
Export to Storage	indefinite (your retention rule)	¢	Manual access

Rule: "Compliance: keep N years, occasional access" → set interactive for the hot window + archive for the rest. **12 years is the archive max.** If you need > 12 years → export to Storage.

🌳 Tree 6: Security posture — Defender for Cloud vs Sentinel

```
"At-a-glance score / posture / recommendations across hybrid + multi-cloud"
  → Microsoft DEFENDER FOR CLOUD (CSPM + CWPP)

"SIEM / SOAR / collect logs from many sources / KQL hunt / run playbook
 on alert / orchestrate response"
  → Microsoft SENTINEL

They overlap: Sentinel can ingest Defender alerts. The verb tells you which
to pick — score → Defender; hunt/respond → Sentinel.

NOT this trio: Microsoft PURVIEW = govern/classify the DATA itself
(catalog, lineage, sensitivity). Out of AZ-305 scope (DP/SC exams).
Sentinel watches traffic · Defender scores posture · Purview inventories cargo.
```

Alerts: metric vs log vs activity-log vs smart-detection

Trigger	Use
Threshold on a numeric metric (CPU > 80%)	**Metric alert**
KQL query over Log Analytics data	**Log alert**
Subscription-level event (resource created, RBAC change)	**Activity log alert**
Health degradation (region/service down)	**Resource Health alert**
ML-detected anomaly in App Insights	**Smart Detection**

D2 — Data Storage

🌳 Tree 1: SQL family selection

```
Need SQL Server compatibility level?
├─ Need OS-level access, SQL Agent across DBs, FileStream, cross-DB queries,
│  CLR, Service Broker, SSAS/SSRS on the same box, full TDE BYOK with HSM
│  customer-managed key on storage → SQL ON AZURE VM (IaaS)
│
├─ Need MOST of SQL Server features (cross-DB queries, SQL Agent, Service
│  Broker, DB Mail, CLR, linked servers) but want PaaS managed updates +
│  VNet integration → AZURE SQL MANAGED INSTANCE
│  (lives in a DEDICATED SUBNET — delegated, min /27, no other resources)
│
└─ Single-database / multi-tenant SaaS / no cross-DB queries needed,
   maximum PaaS abstraction → AZURE SQL DATABASE
```

Lift-and-shift lock: SQL MI for migrating SQL Server Enterprise / Always On AG estates (instance-scoped, near-100% compatibility). **Azure SQL DB** for new or refactored cloud-native apps.

🌳 Tree 2: Azure SQL DB tier selection

```
Purchasing model?
├─ Simple, pay-per-DTU bundles (compute+storage+IO) → DTU MODEL
│   ├─ Basic   (5 DTU)        – dev/test only
│   ├─ Standard (S0..S12)     – most OLTP. Columnstore requires S3+
│   └─ Premium (P1..P15)      – high IO, in-memory OLTP, sub-2ms,
│                               read scale-out
│
└─ Independent compute + storage scaling, transparency, reservations
   → vCORE MODEL
   ├─ General Purpose         – most workloads (remote storage, OK perf)
   ├─ Business Critical       – local SSD, in-memory OLTP, AG-based
   │                            HA + 1 free readable replica
   └─ Hyperscale              – >4 TB databases, near-instant backups,
                                up to 4 readable replicas, fast scale-out
```

DTU sizing rule: estimate the DTU floor from IOPS (rough rule: workload IOPS divided by 3). Round **up** to the smallest tier ≥ floor. Don't jump to Premium unless the question specifies in-memory OLTP, sub-2ms latency, or read scale-out.

Hyperscale trigger phrases: "100 TB", "fast restore for very large DB", "near-instant scaling".

🌳 Tree 3: Cosmos DB API selection

```
What data shape + existing client SDK do you have?
├─ Document (JSON), greenfield → NoSQL API (a.k.a. SQL API, "Core")
│  – Microsoft's flagship, all newest features land here first
├─ Existing MongoDB app                              → MongoDB API
├─ Existing Cassandra app                            → Cassandra API
├─ Existing Gremlin graph app                        → Gremlin API
├─ Existing Azure Table Storage app, want global +   → Table API
│  better SLA + autoscale
└─ PostgreSQL workload that needs horizontal sharding → Cosmos for PostgreSQL
   across many nodes (Citus under the hood)            (separate product)
```

🌳 Tree 4: Cosmos DB consistency level

```
"Strongest possible, never see stale data, any latency cost OK"
                                                      → STRONG

"Bounded lag in time OR versions; recent reads;
 cheaper than Strong; in-region OK"                   → BOUNDED STALENESS

"User reads their own writes within their session
 (typical web app per-user pattern)" – DEFAULT        → SESSION

"Reads always in write-order, but freshness unbounded"  → CONSISTENT PREFIX

"Eventually consistent, no order guarantee, cheapest,
 lowest latency"                                      → EVENTUAL
```

Strength order (strong → weak): Strong > Bounded staleness > Session > Consistent prefix > Eventual. **Default is Session. Multi-region writes require Session or weaker** (Strong with multi-write is supported only in specific configurations — exam usually says "no, weaken to bounded staleness if you need multi-write").

🌳 Tree 5: Storage account types

```
General-purpose, all blob tiers, modern features              → StorageV2 (default)
Block blobs + page blobs only, classic                        → StorageV1 (legacy, don't pick)
Very-high-throughput block blob workloads, low-latency        → BlockBlobStorage (Premium)
ADLS Gen2 (HNS = hierarchical namespace), big data analytics  → StorageV2 with HNS enabled
File shares with NFS/SMB at low latency                       → FileStorage (Premium)
```

🌳 Tree 6: Blob access tier

Tier	Min duration	Use when
Premium	n/a	Block blob, ultra-low latency, transactional
Hot	none	Frequently accessed
Cool	**30 days**	Infrequent access, accessible immediately
Cold	**90 days**	Rarely accessed (~90d+), accessible immediately, cheaper than Cool
Archive	**180 days**	Rarely accessed, **rehydration delay (hours)**

Trigger lock: "Need immediate access" → never Archive. "Rehydration acceptable" → Archive.

🌳 Tree 6b: Blob lifecycle management (automatic tier transitions)

What it is: a JSON policy on the storage account that automatically **moves blobs between tiers** or **deletes them** based on age / last access / last modification — no code, no Functions, no Logic Apps required.

```
Requirement
├─ "Move blobs to Cool after 30 days, Archive after 180, delete after 7 yrs,
│   no custom code"                                    → LIFECYCLE MANAGEMENT POLICY
│   (rules on storage account, runs daily, free)
│
├─ "Tier based on last ACCESS time (not last modified)"   → enable LAST ACCESS
│   TIME TRACKING first, then `daysAfterLastAccessTimeGreaterThan` in policy
│
├─ "Need to react the instant a blob is created/modified" → EVENT GRID
│   (not lifecycle — lifecycle runs ~once/day)
│
└─ "Custom business logic per blob (e.g., based on metadata/content)"
                                                      → AZURE FUNCTION + Event Grid
```

Hard rules on policies (exam trap territory):

- Policy runs **once per day**, can take **up to 24-48 h** to take effect.
- Applies to **block blobs and append blobs** only (not page blobs / disks).
- Moves are **one-way down the temperature ladder** by default: Hot → Cool → Cold → Archive. Going **back up** = explicit rehydration (cost + delay).
- **Early-deletion penalty:** if you move/delete before the tier minimum (Cool 30 d, Cold 90 d, Archive 180 d), you're billed for the remaining days.
- Filters: prefix match, blob index tags, blob type. **No regex.**
- Conditions: `daysAfterModificationGreaterThan`, `daysAfterLastAccessTimeGreaterThan`, `daysAfterCreationGreaterThan` (snapshots/versions only).

Trigger phrases: "automatically tier", "based on age / last access", "no admin effort", "scheduled archival", "delete old data after N days" → **lifecycle management policy**. Not Functions, not Data Factory.

🌳 Tree 7: Storage redundancy

Redundancy	Replicas	Survives	Read access on secondary?
LRS	3 copies in 1 datacenter	Disk failure	—
ZRS	3 copies across 3 AZs in 1 region	1 AZ failure	—
GRS	LRS + async copy to paired region	Region failure	No (only after failover)
GZRS	ZRS + async copy to paired region	Region + AZ failure	No
RA-GRS	GRS + read on secondary endpoint	Region failure	**Yes (read-only)**
RA-GZRS	GZRS + read on secondary endpoint	Region + AZ failure	**Yes (read-only)**

Trigger lock: "Multi-region read access without failover" → RA-GRS or RA-GZRS. "Survive a zone failure" → ZRS / GZRS / RA-GZRS.

Redundancy ladder: LRS < ZRS < GRS < GZRS. Add the `RA-` prefix for a read-access secondary. When **both zone and geo** redundancy are required, always pick the **GZRS family** (GZRS / RA-GZRS).

🌳 Tree 8: Data ingestion / integration

```
Source / pattern
├─ Scheduled batch ETL from many sources (DB→DB, file→DB, API→DW)  → DATA FACTORY
│  • Also valid: Synapse Pipelines (same engine, in Synapse workspace)
│
├─ Streaming events (IoT, telemetry) needing windowed aggregation  → STREAM ANALYTICS
│  on Event Hubs / IoT Hub input
│
├─ React to a discrete resource event (blob created, RG modified,  → EVENT GRID
│  Key Vault secret expired) — push, low-latency, serverless
│
├─ High-throughput append-only event stream (millions/sec),        → EVENT HUBS
│  pull-based consumer, Kafka-compatible
│
└─ Enterprise messaging with FIFO, dead-letter, transactions,      → SERVICE BUS
   sessions, large message size (1 MB std / 100 MB Premium)
```

Synapse compute options

- **Dedicated SQL pool** = MPP data warehouse, pre-provisioned DWUs
- **Serverless SQL pool** = ad-hoc T-SQL over data lake files, pay-per-TB-scanned
- **Spark pool** = notebooks, ML, big data processing
- **Synapse Link** = near-real-time replication from Cosmos DB / SQL / Dataverse into the analytical store (no ETL)

D3 — Business Continuity

🌳 Tree 1: Backup vs DR (basics)

```
Question is about RECOVERING DATA (point in time, ransomware,
accidental delete)                                          → BACKUP

Question is about KEEPING SERVICE RUNNING during a regional /
zonal failure                                               → DR / HA

RPO (Recovery Point Objective)  = how much DATA can you afford to lose?
                                  (gap between backups / replication)
RTO (Recovery Time Objective)   = how long can you afford to be down?
                                  (time to bring service back)

Smaller RPO/RTO = more expensive / more complex.
```

🌳 Tree 2: Backup Vault vs Recovery Services Vault

```
What are you backing up?
├─ VMs, on-prem (MARS/MABS), Azure Files, SQL in VM, SAP HANA in VM,
│  Azure Backup Server                                      → RECOVERY SERVICES VAULT
│  (the older, broader vault; "classic" workloads)
│
└─ Azure Blobs, Azure Disks, Azure Database for PostgreSQL Flex,
   AKS, Azure Kubernetes Service                            → BACKUP VAULT
   (newer; modern PaaS-native workloads)

Both managed under AZURE BACKUP CENTER (single pane of glass).

Soft delete on either vault: 14 days default, up to 180 days max.
"Enhanced soft delete" = always-on, can't be disabled (modern default).
```

🌳 Tree 2b: Recovery after deletion (per workload)

Mental model: there are **3 layers of protection** against accidental deletion. Pick based on what was deleted and how far back you need to go.

```
Layer 1 — PREVENT deletion            → resource LOCKS (CanNotDelete / ReadOnly)
Layer 2 — UNDO recent deletion        → SOFT DELETE / VERSIONING / PITR
Layer 3 — RESTORE from a backup vault → Recovery Services / Backup Vault
```

Per-resource lookup table:

Deleted thing	Recovery mechanism	Default window	Max window
Blob / blob version	Blob soft delete	7 days	**365 days**
Blob container	Container soft delete	7 days	365 days
Blob (overwritten, not deleted)	**Blob versioning**	unlimited until deleted	—
Whole storage account (block blobs)	**Point-in-time restore (PITR)** — needs versioning + change feed + soft delete	up to 365 days	365 days
Azure Files share content	Share + file soft delete + share snapshots	7 days (soft delete)	365 days
Managed disk	Disk soft delete (subscription setting)	7 days	14 days; or Backup Vault for longer
Key Vault / secret / key / cert	Soft delete + **purge protection**	90 days (fixed)	90 days
Resource group / resource	None natively — use **resource lock** to prevent, or restore from Backup Vault / ARM deployment redeploy		
Entra ID user / group	Recycle bin	30 days (hard)	30 days
Azure SQL DB	**Point-in-time restore (PITR)**	**7 days** (default), up to **35 days**	35 days; **LTR up to 10 yrs**
Azure SQL MI	PITR + LTR	same as SQL DB	35 d / 10 yrs
Cosmos DB	**Continuous backup (PITR)** Tier 7 or **Tier 30**	30 days max	30 days; or periodic backups
VM	RSV restore (snapshot-based)	per backup policy	per policy
Backup item in vault	**Vault soft delete**	**14 days**	180 days; Enhanced soft delete on by default

Key trap distinctions (exam loves these):

- **Soft delete ≠ versioning.** Soft delete = recover after **delete**. Versioning = recover after **overwrite**. Real protection = **enable both**.
- **PITR for blobs requires three things on**: versioning, change feed, and blob soft delete. Question saying "enable PITR" implies all three.
- **Purge protection (Key Vault):** once on, **cannot be turned off**, and soft-deleted items **cannot be purged** before the retention period — this is mandatory for compliance (PCI, FedRAMP, etc.).
- **Resource locks DON'T protect data inside** — a CanNotDelete lock on a storage account stops account deletion but **doesn't stop blob deletion**.
- **Recycle bin (Entra ID)** is 30 days **fixed** — non-configurable.
- **Immutable blob storage (WORM)** is different from soft delete: it **blocks deletion entirely** for the retention period (legal hold / time-based). Pick this for "regulatory / can't be deleted even by admin". **WORM hierarchy (weakest → strongest):** legal hold (removable, indefinite, ad-hoc) < time-based retention (configured, time-bounded) < **locked** time-based retention (immutable, SEC 17a-4 compliant — what regulators expect).

Trigger phrases:

- "Accidentally deleted blob, recover" → blob soft delete (if within window)
- "Recover overwritten blob to previous state" → blob versioning
- "Restore storage account to point in time" → PITR (needs all 3 prereqs)
- "Recover SQL DB to specific moment in last week" → PITR
- "Restore SQL DB from 6 months ago" → LTR (Long-Term Retention)
- "Prevent accidental deletion" → resource lock + soft delete + purge protection
- "Must be unable to delete for 7 years (regulation)" → immutable storage (WORM) or LTR

🌳 Tree 3: Disaster Recovery — multi-region pattern

```
Latency / cost / RTO / RPO trade-off ladder (cheapest → most expensive):

BACKUP & RESTORE
├─ RTO: hours-days, RPO: depends on backup cadence
├─ No standing compute in secondary
└─ "Restore on disaster" — cheapest, slowest

PILOT LIGHT
├─ RTO: hours, RPO: minutes (replicated data)
├─ Critical data replicated; compute provisioned at failover (IaC)
└─ Cheap-ish, needs DR drills

WARM STANDBY
├─ RTO: minutes (~5-15 min typical), RPO: seconds-minutes
├─ Secondary runs at MIN scale, autoscales out on failover
├─ Often paired with Traffic Manager priority routing
└─ Moderate cost; needs scale-up time during failover

ACTIVE-ACTIVE (HOT-HOT)
├─ RTO: seconds, RPO: ~zero (sync or near-sync replication)
├─ BOTH regions serve traffic; no scale-up needed
├─ Front Door / Traffic Manager weighted/performance routing
└─ Most expensive; mission-critical only (banks, payment APIs)
```

Trigger phrases: "must serve from both regions" → active-active. "RTO < 60s" → active-active. "Can take 15 min to fail over, cheaper" → warm standby. "Restore acceptable in 4 hours" → pilot light or backup.

🌳 Tree 4: Azure Site Recovery (ASR) scope

```
ASR replicates and orchestrates failover for:
├─ Azure VMs between Azure regions                    ✅
├─ On-prem VMware / Hyper-V / physical servers TO Azure  ✅
├─ Azure Stack VMs TO Azure                           ✅
├─ Replication FROM Azure public MEC TO region        ✅
├─ Replication BETWEEN two Azure public MECs          ✅
├─ Azure SQL Managed Instance across regions          ❌ (use Failover Groups)
└─ PaaS services (App Service, SQL DB, Cosmos)        ❌ (use the service's built-in geo)

ASR is ACTIVE-PASSIVE by design (not active-active).
"DRaaS in Azure" = ASR. Always.
```

🌳 Tree 5: SQL HA inside one region

Need	Use
Auto-failover within region, **fully managed**	**Always On Availability Group** (built-in to SQL MI Business Critical, and to SQL DB Business Critical / Premium)
Shared-storage cluster on VMs	**Failover Cluster Instance (FCI)** on SQL on Azure VM
Async log delivery for warm read-only secondary	**Log shipping** (legacy, exam will rarely pick)
Read scale-out, no failover	**Active geo-replication** (SQL DB) or **read replicas** (Hyperscale)
Cross-region auto-failover for SQL DB / MI	**Failover Groups** (wrap geo-replication + DNS-listener swap)

Cosmos DB multi-region

- Single-write region + multiple read regions → default geo-replication (RPO ~seconds).
- **Multi-write regions** (multi-master) → optional. Trades off consistency (cannot use Strong with multi-write outside specific configs).
- Failover: **automatic** (Microsoft) or **manual** (you trigger). For RTO guarantees, configure automatic failover priority order.

D4 — Infrastructure (Compute, Networking, Migrations)

🌳 Tree 1: Compute service selection

```
Step 1: Code, container, or VM?

├─ I HAVE A VM IMAGE / NEED OS CONTROL
│   ├─ Single, stateful workload                              → AZURE VM
│   ├─ Identical scale-out fleet, autoscale                   → VMSS
│   │   (Flexible orchestration = mix sizes/AZ-spread;
│   │    Uniform = identical, faster scale)
│   ├─ HPC / MPI / tightly-coupled parallel jobs              → AZURE BATCH
│   └─ Need DR for VMs                                        → ASR (above)
│
├─ I HAVE A CONTAINER
│   ├─ Just need to run ONE container, quickly, no orchestrator → AZURE CONTAINER INSTANCES (ACI)
│   │   (single container group, no scaling, no scale-to-zero)
│   │
│   ├─ Need scale-to-zero + per-request billing + HTTP/event    → AZURE CONTAINER APPS (ACA)
│   │   triggers + VNet + KEDA built-in
│   │   Linux containers ONLY (linux/amd64). Windows web tier → App Service (Windows).
│   │
│   ├─ Need full Kubernetes (service mesh, custom controllers,  → AZURE KUBERNETES SERVICE (AKS)
│   │   DaemonSets, multi-team multi-tenant)
│   │   Sub-trees: Automatic (managed) vs Standard. Cluster
│   │   autoscaler + KEDA + node pools for spot / GPU / Windows.
│   │
│   └─ App Service supports containers too                     → APP SERVICE FOR CONTAINERS
│       (always-on plan, no scale-to-zero, easiest if already on App Service)
│
└─ I HAVE CODE
    ├─ Web app / API, always-on, simple deploy                 → APP SERVICE (Linux/Windows)
    │   (Plans: F/D free, B basic, S standard, P/Pv2/Pv3
    │    production, I/Iv2 isolated/ASE)
    │
    ├─ Event-driven, ephemeral, scale-to-zero                  → AZURE FUNCTIONS
    │   ├─ Consumption plan       → scale-to-zero, pay-per-exec, 5-min cap
    │   ├─ Premium plan           → pre-warmed (no cold start), VNet, longer runs
    │   └─ Dedicated (App Service) → run on existing ASP, no scale-to-zero
    │
    └─ Workflow orchestration / SaaS connectors / no code       → LOGIC APPS
```

Scale-to-zero quick lookup

Service	Scale to zero?
Azure Container Apps	✅ (min replicas = 0)
Azure Functions Consumption	✅
Azure Functions Premium	❌ (always-warm min)
App Service Plans (any tier)	❌
AKS with KEDA	✅ (workload-level)
ACI	n/a (you start/stop it manually)

🌳 Tree 2: Load balancer selection

```
What layer? Where?

GLOBAL (multi-region) + HTTP/HTTPS                              → AZURE FRONT DOOR
  (L7, CDN, WAF, path/host routing, TLS termination at edge,
   anycast, supports private origins via Private Link)

GLOBAL (multi-region) + non-HTTP (TCP/UDP/AMQP/SMTP)            → TRAFFIC MANAGER
  (DNS-based, L4, returns endpoint IP; client connects directly)

REGIONAL + HTTP/HTTPS + path/host routing, WAF, SSL offload,    → APPLICATION GATEWAY
  cookie affinity, in-VNet                                        (L7)

REGIONAL + any TCP/UDP, ultra-low latency, in-VNet               → AZURE LOAD BALANCER
                                                                   (L4; Standard SKU
                                                                    for zone-redundant)
```

Decision shortcut: Global + HTTP = Front Door. Global + non-HTTP = Traffic Manager. Regional + HTTP = App Gateway. Regional + anything else = Load Balancer.

Traffic Manager — 6 routing methods

(See mnemonic in appendix.) **Priority, Weighted, Performance, Geographic, Multivalue, Subnet.** "Security groups" is **NOT** a routing method.

Front Door SKUs

- **Front Door Classic** — legacy.
- **Front Door Standard** — CDN + LB + basic WAF managed rules.
- **Front Door Premium** — Standard + advanced WAF (bot, custom rules) + Private Link to origins + advanced analytics.

🌳 Tree 3: VNet topology

```
Small / mid: hub-spoke with VNet peering — most common AZ-305 answer.
  ├─ Hub: shared services (Firewall, Bastion, VPN/ER Gateway, DNS)
  ├─ Spokes: workloads, peered to hub (non-transitive peering)
  └─ Need transitive between spokes? → route via NVA/Azure Firewall in hub

Large enterprise multi-region, many on-prem branches, SD-WAN     → VIRTUAL WAN
                                                                   (managed hub,
                                                                    transit, ER + VPN
                                                                    + P2S aggregation,
                                                                    secure hub w/ FW)

Cross-subscription peering                                       ✅ supported
Cross-tenant peering                                             ✅ supported (with permission)
Cross-region peering                                             ✅ Global VNet Peering
```

🌳 Tree 4: On-prem ↔ Azure connectivity

```
Bandwidth / SLA / latency need?

≤ 1 Gbps, internet OK, encrypted IPsec                          → SITE-TO-SITE VPN
  (VPN Gateway: VpnGw1..5, max ~10 Gbps aggregate Gen2 SKUs)

Individual remote user (laptop)                                  → POINT-TO-SITE VPN
  (OpenVPN / IKEv2 / SSTP)

Dedicated, private (no public internet), 50 Mbps – 100 Gbps,    → EXPRESSROUTE
  99.95% SLA, ultra-low latency, BGP                               (via connectivity
                                                                    partner / direct)

ER + VPN backup for resilience                                   → ExpressRoute + VPN
                                                                   failover (Microsoft
                                                                   recommends this for
                                                                   mission-critical)

Many branches with SD-WAN                                        → VIRTUAL WAN
                                                                   (aggregates ER + VPN)
```

🌳 Tree 5: DDoS Protection tier

Tier	Cost	Coverage	When
Infrastructure	Free, default	Microsoft's global L3/L4 mitigation, no telemetry to you	Default everyone has
IP Protection	Per-IP/month	Per public IP: telemetry, attack reports, no DRR team, no WAF discount	Small footprint (1-10 IPs), startups, cost-sensitive
Network Protection	Flat tenant fee (~$2,944/mo)	Covers ALL public IPs in protected VNets, DRR team access, WAF discount, cost guarantee	Enterprises with many public IPs

L3/L4 only. For L7 (HTTP request flood) → **Application Gateway WAF** or **Front Door WAF**.

🌳 Tree 6: SQL Managed Instance subnet

- **Dedicated subnet** (no other resources).
- **Delegated** to `Microsoft.Sql/managedInstances`.
- Minimum size: **/27** (32 IPs).
- NSG rules required for management traffic (Microsoft documents the list).
- Cannot share subnet with VMs, AKS, etc.

🌳 Tree 7: Migration ("which R?")

```
Workload's destiny?

REHOST (lift-and-shift)         → Move to Azure VM as-is. Fastest, least
                                  benefit. "Move the data center."

REFACTOR (repackage)             → Minor changes to use PaaS. e.g., move
                                  SQL Server VM to Azure SQL DB. Some
                                  code/config changes.

REARCHITECT                      → Significant restructuring. Monolith →
                                  microservices, add containers, decompose
                                  layers. Major engineering work.

REBUILD                          → Discard the old app; rewrite cloud-native
                                  from scratch (Functions, Cosmos, etc.).

REPLACE                          → Retire the app entirely; adopt SaaS
                                  equivalent (e.g., on-prem CRM → Dynamics 365,
                                  on-prem Exchange → Exchange Online).

RETAIN                           → Keep on-prem for now (compliance,
                                  dependencies, or just not worth it).

RETIRE                           → Decommission the workload; nobody needs it.
                                  (Sometimes listed; sometimes folded into
                                   Retain.)
```

Trigger phrases:

- "Vendor has a SaaS version, we'll subscribe to that" → Replace.
- "Legacy app, just lift to VMs" → Rehost.
- "Move DB to Azure SQL but keep app the same" → Refactor.
- "Rewrite as serverless microservices" → Rebuild.
- "Compliance forbids moving" → Retain.
- "Nobody uses this anymore" → Retire.

🌳 Tree 8: CAF Migrate methodology — 3 phases

```
ASSESS → DEPLOY → RELEASE (also called Optimize)

ASSESS    Discover, inventory, rightsize, cost estimate.
          Tools: Azure Migrate (Discovery and Assessment).

DEPLOY    Replicate, migrate, cutover.
          Tools: Azure Migrate (Server Migration), DMS (Database Migration
          Service), Azure Database Migration, Storage Migration Service.

RELEASE   Stabilize, optimize cost/perf, decommission on-prem.
          Tools: Advisor, Cost Management, Monitor.
```

Don't confuse with the broader CAF lifecycle (Strategy → Plan → Ready → Adopt → Govern → Manage). Govern is NOT a migration phase.

Azure Migrate tools (what's inside)

Tool	Purpose
Discovery and Assessment	Inventory, dependency map, sizing, cost estimate
Server Migration	Replicate + cutover VMware / Hyper-V / physical / AWS / GCP VMs
Database Migration Service (DMS)	SQL Server → Azure SQL / MI / VM, MySQL, PostgreSQL, MongoDB
Web App Migration Assistant	IIS / Tomcat → App Service
Data Box	Offline bulk data ship (TB-PB)
Azure Data Box Gateway / Stack Edge	Hybrid online data transfer

Failure hierarchy (when sizing redundancy)

```
Region (geography, paired regions for GRS)
  └─ Availability Zone (AZ — 3 per zone-enabled region)
       └─ Fault Domain (rack — physical isolation within an AZ/dc)
            └─ Update Domain (patch group — logical isolation for rolling updates)
                 └─ Physical Host (the actual server)
```

Availability Set = FD + UD spread within ONE datacenter (legacy, no AZ protection). **Availability Zones** = spread across AZs (modern; protects against datacenter loss). VMSS Flexible can spread across both.

Mnemonic appendix (arbitrary lists)

These have no internal logic — just memorize.

Traffic Manager — 6 routing methods

"PW PGM S" → **P**riority, **W**eighted, **P**erformance, **G**eographic, **M**ultivalue, **S**ubnet.

NOT a routing method: Security groups, Round-robin (Multivalue is the closest equivalent), Failover (that's the *effect* of Priority).

Cosmos DB — 5 consistency levels (strong → weak)

"Sober Bartenders Serve Cold Eggnog"

Strong → **B**ounded staleness → **S**ession (default) → **C**onsistent prefix → **E**ventual.

Stream Analytics — 5 windowing functions

"Tom Has Sliding Session Snapshots" → **T**umbling (fixed non-overlapping), **H**opping (overlap, size + hop), **S**liding (event-driven thresholds), **S**ession (idle-gap), **S**napshot (same-timestamp grouping).

Cloud migration — the 6 Rs (sometimes 7)

"Rehost, Refactor, Rearchitect, Rebuild, Replace, Retain" (+ optionally Retire = decommission.) Mnemonic: **"Real Refrigerators Really Run Replacing Refrigerators"** — ok that's silly, but the order is roughly **effort low → high**: Retain < Rehost < Refactor < Rearchitect < Rebuild ≈ Replace.

SQL DB DTU Standard ladder (memorize the steps)

S0 (10) → S1 (20) → S2 (50) → **S3 (100)** → S4 (200) → S6 (400) → S7 (800) → S9 (1600) → S12 (3000).

Columnstore = S3 minimum. Premium = P1 (125) … P15 (4000).

Blob tier minimum retention (penalty if you delete early)

- Hot: none
- Cool: **30 days**
- Cold: **90 days**
- Archive: **180 days**

Soft delete defaults (both vaults + storage blob)

- RSV / Backup Vault: **14 days default, up to 180 days max** (Enhanced soft delete on by default on modern vaults — can't be disabled).
- Blob / container soft delete: **7 days default, up to 365 days max**.
- Key Vault soft delete: **90 days fixed** (+ optional purge protection, irreversible once enabled).
- Entra ID recycle bin (deleted users): **30 days fixed**.
- Managed disk soft delete: **7 days default, up to 14 days max**.

CAF lifecycle (whole framework)

"Stay Plenty Ready And Govern Methodically" → **S**trategy → **P**lan → **R**eady → **A**dopt → **G**overn → **M**anage.

Migrate fits inside **Adopt**. Migrate's own 3 phases: Assess → Deploy → Release.

Entra ID — what's in each tier (quick recall)

- **Free:** SSO, basic MFA, security defaults
- **P1: Conditional Access**, SSPR writeback, App Proxy, dynamic groups, Connect Health
- **P2: PIM**, **Identity Protection** (risk-based)
- **Entra ID Governance** (add-on): Access Reviews (modern), Entitlement Management, Lifecycle Workflows, PIM for Groups

Vault → workload quick map

Workload	Vault
Azure VM	RSV
On-prem (MARS / MABS / DPM)	RSV
Azure Files	RSV
SQL in Azure VM	RSV
SAP HANA in Azure VM	RSV
Azure Blob	**Backup Vault**
Azure Disk	**Backup Vault**
AKS	**Backup Vault**
PostgreSQL Flexible Server	**Backup Vault**

Both managed via **Backup Center**.

Front Door vs Traffic Manager vs App Gateway vs LB (decision crib)

Need	Pick
Global, HTTP/S, WAF, CDN	Front Door
Global, any protocol, DNS-based	Traffic Manager
Regional, HTTP/S, WAF, in-VNet	App Gateway
Regional, any TCP/UDP, in-VNet	Load Balancer

Event Hubs vs Service Bus vs Event Grid

Pattern	Service
Pull, massive append-only stream (millions/sec)	**Event Hubs**
Pull, enterprise messaging (FIFO, sessions, dead-letter)	**Service Bus**
Push, reactive to discrete events (blob created, etc.)	**Event Grid**

Exam-day logistics

Before exam

- ✅ Sleep > extra quiz pass. Don't cram.
- ✅ Hydrate, eat normal breakfast, normal coffee.
- ✅ Bathroom **immediately before** start.
- ✅ ID ready (passport or government photo ID + secondary).
- ✅ Quiet room, second monitor unplugged, no other devices in view.
- ✅ Webcam clear of background notes.

During exam

- ⏱️ **~1.7 min per question average** — don't rush, don't stall.
- 🚩 **Flag-and-return** liberally; first instinct usually right.
- ❌ **Never leave blank** — wrong guesses don't penalize, blanks score 0.
- 📖 Read the **last sentence first** on long case studies — it usually contains the actual question.
- 🎯 Run the decision trees in your head — they make the answer obvious.
- ⚠️ Watch for trap words: "least", "most", "minimum", "maximum", "EXCEPT", "NOT". They flip the right answer.
- 🧘 One hard question ≠ hard exam. Park it, move on, come back.

Mindset

You've scored **87–92%** on every recent mock. Pass = **70%**. You have **~17 percentage points of headroom**. Trust the prep.

Vas a romperla. 🎯🏆